Draft: the finite excess in transfinite nim multiplication

a9lim

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Abstract

Conway's field On_2 makes the ordinals below ω^{ω^ω} an algebraic closure of $\mathbb{F}_2$ under nim-arithmetic. Its multiplication is governed by a Kummer tower whose carries are Lenstra's *excess* values: for each odd prime p, the carry is $\alpha_p = \kappa_{f(p)} + m_p$, where the transfinite part $\kappa_{f(p)}$ has a known closed shape (Lenstra, DiMuro) but the finite correction m_p does not. This note consolidates the state of the excess problem (`OPEN.md` Problem 3 of `ogdoad`): the exact finite-field reformulation of the root test as a multiplicative-order criterion and its structural norm reduction; a complete analysis of the 3-power column via a half-angle splitting in the cyclotomic component fields, certifying $m_r = 1$ for all primes r with $f(r) \in \{3, 9, \ldots, 3^6\}$ and forcing $m_p \geq 4$ unconditionally on the $f(p) = 2 \cdot 3^k$ column — now sharpened to $m_p = 4$ *exactly* at every prime current factor tables reach (universally for $k \leq 6$), via a corrected compositum norm $(\kappa + 4)(\kappa + 5) = \kappa^2 + \kappa + \omega$ that collapses the $m = 4$ test into the same trinomial field; the empirically exact 0/1/4 candidate rule and the proof that the Lenstra component set alone cannot decide the excess; and the reduction of Lenstra's open boundedness question to that candidate. Feasibility of the first open row, $p = 719$, is assessed: a Frobenius-orbit norm recurrence reduces it to arithmetic in $\mathbb{F}_{2^{359}}$, with the cost concentrated in tower-aware Frobenius composition. Each statement carries an explicit claim-level tag.

Claim levels. [**proved**] – unconditional, machine-checked or classically verified; [**certified** $k \leq 6$] – proved for the stated range, blocked beyond it only by unfactored cofactors; [**consistent**] – no counterexample found, not proved; [**conjectured**] – explicitly open in the literature or posed in this thread; [**open**] – neither confirmed nor refuted, no feasible attack on record.

1 Notation, implemented state, external data

1.1 Notation

For an odd prime p:

- $f(p) = \mathrm{ord}_p(2)$, the multiplicative order of 2 modulo p;
- κ_h is the Lenstra/DiMuro tower element indexed by h;
- $\mathcal{Q}(h)$ is Lenstra's set of prime-power components appearing in κ_h;
- the *excess* m_p is the least finite m such that $\kappa_{f(p)} + m$ has no p-th root in the relevant finite component field;
- the Kummer carry is $\alpha_p = \kappa_{f(p)} + m_p$.

For the ordinal tower, a row with $\mathcal{Q}(f(p)) = \{q\}$ and finite excess m gives the ordinal sum corresponding to $\kappa_q + m$. Addition of nimbers is XOR; $\otimes$ is the nim product.

1.2 Implemented tower state

In the Rust tower (`src/scalar/big/ordinal/tower.rs`):

- DiMuro Table 1 rows [3] through α_{43};
- a locally verified row $\alpha_{47} = \omega^{(\omega^7)} + 1$ (Section 7);
- the operational boundary is α_{53}: a carry needing α_{53} or beyond returns `None`.

1.3 External data snapshot (2026-06-09)

- OEIS A380496 [7] has 1417 extended rows: 799 known and 618 unknown; the b-file has 126 initial known rows.
- The first OEIS unknown is row $n = 127$, the 127th odd prime $p = 719$. For $p = 719$: $f(719) = \mathrm{ord}_{719}(2) = 359$ and $\mathcal{Q}(359) = \{359\}$.
- The transfinite-nim-calculator logs [9] record the direct component exponent as $e_{719} = 1{,}258{,}230{,}380$, the practical wall for direct exponentiation.

2 The order criterion and structural norm reduction

2.1 The exact criterion

Throughout, E denotes the degree of the component field containing $\beta = \kappa_{f(p)} + m$, so $f = f(p) \mid E$ and hence $p \mid 2^E - 1$.

Proposition 1 (power criterion)**.** *[**proved**] $\beta \in \mathbb{F}_{2^E}^\times$ has no p-th root in $\mathbb{F}_{2^E}$ iff $\beta^{(2^E-1)/p} \neq 1$. Hence*

$$m_p = \min\{m \geq 0 : \ (\kappa_{f(p)} + m)^{(2^E-1)/p} \neq 1\}.$$

Proof. $\mathbb{F}_{2^E}^\times$ is cyclic of order $2^E - 1$ and $p \mid 2^E - 1$; in a cyclic group of order N, β is a p-th power iff $\beta^{N/p} = 1$. □

Corollary 1 (order form)**.** *[**proved**] If $v_p(2^E - 1) = 1$, then β has no p-th root in $\mathbb{F}_{2^E}$ iff $p \mid \mathrm{ord}(\beta)$, so m_p is the least m with $p \mid \mathrm{ord}(\kappa_{f(p)} + m)$.*

Remark 1 (Wieferich caveat)**.** In general "no p-th root" is equivalent to $v_p(\mathrm{ord}\,\beta) = v_p(2^E - 1)$, which collapses to $p \mid \mathrm{ord}(\beta)$ only when $v_p(2^E - 1) = 1$. Since $v_p(2^E - 1) = v_p(2^{f(p)} - 1)$ whenever $p \nmid E/f(p)$ (lifting the exponent), the hypothesis can fail only at base-2 Wieferich-type primes with $p^2 \mid 2^{f(p)} - 1$; the only known instances are $p = 1093$ and $p = 3511$, both of which lie inside the extended A380496 range. For those rows the exact test of Proposition 1 (or the full-p-part order condition) must be used. The certifications of Section 3 compute orders exactly, with squarefreeness checks, and are unaffected. The research-note predecessor of this document stated only the order form; the power criterion is the correct general statement.

2.2 Norm reduction

Proposition 2 (norm reduction)**.** *[**proved**] For $\beta \in \mathbb{F}_{2^E}^\times$ with $f = f(p) \mid E$,*

$$\beta^{(2^E-1)/p} = \mathrm{N}_{\mathbb{F}_{2^E}/\mathbb{F}_{2^f}}(\beta)^{(2^f-1)/p}.$$

Proof. $\mathrm{N}_{\mathbb{F}_{2^E}/\mathbb{F}_{2^f}}(\beta) = \beta^{(2^E-1)/(2^f-1)}$, and $\big((2^E-1)/(2^f-1)\big)\cdot\big((2^f-1)/p\big) = (2^E-1)/p$. $\square$

So the $p = 719$ test reduces to: (1) compute $N = \mathrm{N}_{\mathbb{F}_{2^E}/\mathbb{F}_{2^{359}}}(\kappa_{359}+1)$ in $\mathbb{F}_{2^{359}}$; (2) test whether $719 \mid \mathrm{ord}(N)$ in $\mathbb{F}_{2^{359}}^{\times}$, a field of degree 359 in which $719 \mid 2^{359}-1$.

Proposition 3 (Galois-conjugacy invariance)**.** *[***proved***] The order of $\kappa_{f(p)}+m$ is constant on its Frobenius orbit in $\mathbb{F}_{2^E}$, and the norm $N = \prod_{i=0}^{E/f-1}\mathrm{Frob}^{if}(\kappa_f+m)$ is determined structurally by the tower equations without materialising the full orbit.*

Proposition 4 (tower pinning)**.** *[***proved***] The tower equations pin κ_{359} up to Frobenius conjugacy: each composed minimal polynomial along the component chain $359 \to 179 \to 89 \to 11 \to 5 \to$ finite is irreducible (the non-q-th-power property defining the lower excess guarantees this), so κ_{359} is a well-defined conjugacy class and N is a well-defined element of $\mathbb{F}_{2^{359}}$.*

2.3 Feasibility for $p = 719$

The component field degree is $E = 2\cdot 2\cdot 5\cdot 11\cdot 89\cdot 179\cdot 359 = 1{,}258{,}230{,}380$; the norm requires $E/f = 3{,}504{,}820$ Frobenius steps.

[**consistent**] The norm can in principle be computed by the binary-splitting Frobenius recurrence $N_{2k} = N_k\cdot\mathrm{Frob}^{fk}(N_k)$ (iterated-Frobenius technique, von zur Gathen–Shoup [10], Kaltofen–Shoup [4]), requiring about $\log_2(3{,}504{,}820) \approx 22$ squarings-with-Frobenius-composition at degree E over $\mathbb{F}_2$. Each Frobenius composition is a modular composition in $\mathbb{F}_2[x]$ modulo a degree-E polynomial; with dense polynomial arithmetic at $E \approx 1.26\times 10^9$ this is not locally feasible.

[**conjectured**] (as a cost model, not a theorem) Kummer-tower-aware Frobenius – standard-lattice arithmetic for compatibly embedded finite fields in the style of De Feo–Randriam–Rousseau [2], where q-power Frobenius acts near-monomially per tower level – is the likely 10–100× lever: the sparse factored structure $E = 2\cdot 2\cdot 5\cdot 11\cdot 89\cdot 179\cdot 359$ admits a stepwise Frobenius representation in which each level contributes a small-degree modular composition. A *measured* cost model, not a run, is the correct first deliverable.

[**open**] Whether the Kummer-tower Frobenius model makes the m_{719} computation feasible in wall-clock hours rather than months on available hardware (a single consumer GPU; GF(2) arithmetic is XOR-sliced and GPU-friendly).

Remark 2 (shared abstraction)**.** The structural primitive needed here – $\mathrm{N}_{E/K}(\beta) = \prod_i \mathrm{Frob}^i(\beta)$ computed by a Frobenius-orbit recurrence – is the same primitive that the cyclic-algebra Brauer-invariant bridge (Bridge K, `roadmap/TBD.md`) requires for reduced norms over a Frobenius-generated cyclic extension. The existing `FieldExtension::relative_norm` does not apply to the term-algebra/transfinite setting, but the shape is identical; factoring out a reusable `relative_norm_over_frobenius_orbit` is an engineering opportunity. This is not a claim that the bounded `Fpn` norm certifies m_{719}.

3 The 3^k family

Write $\zeta := \kappa_{3^k}$ and $h := 3^k$ throughout this section.

3.1 Structure

Proposition 5 (cyclotomic recognition)**.** *[***proved***] $\zeta^{3^k} = \kappa_2$, which has order 3; hence ζ is a primitive 3^{k+1}-st root of unity. Since 2 is a primitive root modulo 3^{k+1}, the cyclotomic polynomial*

$\Phi_{3^{k+1}}(x) = x^{2h} + x^h + 1$ *is irreducible over* $\mathbb{F}_2$*, the component field is* $\mathbb{F}_2(\zeta) = \mathbb{F}_{2^{2h}}$*, and it carries the index-2 subfield* $L = \mathbb{F}_{2^h}$*.*

Proposition 6 (half-angle splitting)**.** *[**proved**] With* $s = (3^{k+1}+1)/2$ *(the inverse of* 2 *modulo* 3^{k+1}*),*

$$\kappa_{3^k} + 1 = \zeta^s \left(\zeta^s + \zeta^{-s}\right),$$

where ζ^s *lies in the norm-one circle* U *(of order exactly* 3^{k+1} *within the subgroup of order* 2^h+1*) and* $(\zeta^s + \zeta^{-s})^2 = \zeta + \zeta^{-1} =: \gamma_k \in L^\times$*. Since* $\mathbb{F}_{2^{2h}}^\times = L^\times \times U$ *with coprime orders,*

$$\operatorname{ord}(\kappa_{3^k} + 1) = 3^{k+1} \cdot \operatorname{ord}(\gamma_k), \qquad \operatorname{ord}(\gamma_k) \mid 2^{3^k} - 1.$$

Machine-checked corollaries: $(\kappa+1)^{2^h-1} = \zeta^{-1}$ *and* $\mathrm{N}_{F/L}(\kappa+1) = \gamma_k$*.*

Proposition 7 (translates $m = 2, 3$)**.** *[**proved**]* $\kappa_{3^k} + 2$ *and* $\kappa_{3^k} + 3$ *split as in Proposition 6 with* γ_k *replaced by a Galois conjugate. Therefore* $\operatorname{ord}(\kappa_{3^k} + m) = 3^{k+1} \cdot \operatorname{ord}(\gamma_k)$ *for* $m = 1, 2, 3$*, and* $\operatorname{ord}(\kappa_{3^k}) = 3^{k+1}$*.*

Proposition 8 (the $2 \cdot 3^k$ exception, unconditionally)**.** *[**proved**] Any prime* p *with* $f(p) = 2 \cdot 3^k$ *satisfies* $p \mid 2^{3^k} + 1$*, hence* p *divides neither* 3^{k+1} *nor* $2^{3^k} - 1$*, hence* p *never divides* $\operatorname{ord}(\kappa_{3^k} + m)$ *for* $m \in \{0, 1, 2, 3\}$*. By the order criterion,* $m_p \geq 4$*. The full primitivity conjecture* C_k *below is not needed; the splitting alone forces the bound. The argument applies beyond the current tables, e.g.* $p = 87211$*,* $f = 54$*.*

Proposition 9 (the 3-power column)**.** *[**proved**] For a prime* r *with* $f(r) = 3^k$*, the values* $m \in \{0, 2, 3\}$ *are impossible:* $m_r = 0$ *would require* $r \mid \operatorname{ord}(\kappa_{3^k}) = 3^{k+1}$*, a power of* 3*, while* $r \equiv 1 \pmod{3^k}$ *is a prime distinct from* 3*; and* $m = 2, 3$ *share* $m = 1$*'s order class by Proposition 7. Any failure of* C_k *therefore gives* $m_r \geq 4$ *directly.*

Proposition 10 (norm tower)**.** *[**proved**]* $\gamma_{k-1} = \gamma_k^3 + \gamma_k = \mathrm{N}_{L_k/L_{k-1}}(\gamma_k)$ *(minimal polynomial* $X^3 + X + \gamma_{k-1}$*), so* $\operatorname{ord}(\gamma_{k-1}) \mid \operatorname{ord}(\gamma_k)$*. With lifting-the-exponent (*$v_r(2^{3^k} - 1) = v_r(2^{f(r)} - 1)$ *for old primes* $r \neq 3$*), full-order parts propagate upward, and conjecture* C_k *reduces to* C_{k-1} *plus primitivity at the new primes* $r \mid \Phi_{3^k}(2)$*.*

Conjecture 1 (C_k: γ-primitivity)**.** γ_k *is primitive in* $L^\times = \mathbb{F}_{2^{3^k}}^\times$*.*

Proposition 11 (equivalence)**.** *[**proved**] For primes* r *with* $f(r) \mid 3^k$ *a 3-power:*

$$C_k \iff m_r = 1 \text{ for every prime } r \text{ with } f(r) \in \{3, 9, \ldots, 3^k\}.$$

The family formula is the 0/1/4 *candidate of Section 4 restricted to the* 3*-power column.*

3.2 Certification status

[**certified** $k \leq 6$] C_k holds for $k = 1, \ldots, 6$: γ_k is primitive in $\mathbb{F}_{2^{3^k}}^\times$, and $\operatorname{ord}(\kappa_{3^k}+1) = 3^{k+1}\,(2^{3^k}-1)$ is verified exactly – product reconstruction, squarefreeness, primality by deterministic Miller–Rabin below 3.3×10^{24} and MR-64 probable-primality above, an independent sieve re-deriving the small factors, cross-validation against the independent term algebra for $k = 1, 2$ and against the calculator rows $m_{2593} = 1$, $m_{487} = 1$ (`experiments/cyclotomic_3k_family.py`, `experiments/excess/`).

[**consistent**] $k = 7, 8$: every known prime factor of $\Phi_{3^7}(2)$ and $\Phi_{3^8}(2)$ divides $\operatorname{ord}(\gamma_k)$. Full certification is blocked only by the unfactored cofactors (FactorDB status composite-no-factor); no failure has been found.

[**open**] Whether ECM or GNFS can factor the $\Phi_{3^7}(2)$ and $\Phi_{3^8}(2)$ cofactors within a realistic budget, converting $k = 7, 8$ from CONSISTENT to CERTIFIED.

3.3 Certified excess rows produced

The following $m_r = 1$ rows are certified by the order criterion independently of the cofactor issue (analysis-level results; they are not new Rust carries):

f	primes r with $m_r = 1$
27	262657
81	71119, 97685839
243	16753783618801, 192971705688577, 3712990163251158343
729	80191, 97687, 379081, P42, P90
2187	39367, 7606246033, 263196614521, 529063556041
6561	209953, 1299079, 7006326739760670927739 3

(P42, P90 denote the certified/probable primes of 42 and 90 decimal digits dividing $\Phi_{3^6}(2)$.)

4 The $0/1/4$ candidate

Conjecture 2 (the 0/1/4 rule)**.** ***[consistent]*** *(not proved)*

$$m_p = \begin{cases} 0 & \textit{if } \mathcal{Q}(f(p)) \textit{ is not a singleton odd prime-power,} \\ 4 & \textit{if } f(p) = 2 \cdot 3^k,\ k \geq 1, \\ 1 & \textit{otherwise (singleton odd prime-power } \mathcal{Q}(f(p))\textit{).} \end{cases}$$

The rule matches all 950 *calculator records with known $\mathcal{Q}$-sets and every OEIS-known row covered by those $\mathcal{Q}$-sets.*

4.1 The component set alone is insufficient

[proved] $\mathcal{Q}(f(p))$ alone does not determine m_p:

$\mathcal{Q}$	$m = 4$	$m = 1$
$\{9\}$	$p = 19,\ f = 18$	$p = 73,\ f = 6$
$\{81\}$	$p = 163,\ f = 162$	$p = 2593,\ f = 18$
$\{243\}$	$p = 1459,\ f = 486$	$p = 487,\ f = 18$

[proved] The split for $\mathcal{Q} = \{9\}$ is explained by the order criterion: $\mathrm{ord}(\kappa_9 + 1) = 3^3 \cdot (2^9 - 1) = 27 \cdot 511 = 13797$ and $511 = 7 \cdot 73$, so $73 \mid \mathrm{ord}(\kappa_9 + 1)$ but $19 \nmid \mathrm{ord}(\kappa_9 + 1)$; adding $m = 4$ changes the order and introduces the factor 19.

[proved] The lower bound of the exception column is unconditional: $m_p \geq 4$ for all p with $f(p) = 2 \cdot 3^k$ (Proposition 8).

[conjectured] The matching upper bound $m_p \leq 4$ on that column; equivalent to C_k for the primes in the exception column.

4.2 The $m = 4$ upper bound: settled at every visible prime

The $m = 4$ translate $\kappa_{3^k} + 4$ involves the number $4 \in \mathbb{F}_{16}$, which does not lie in $\mathbb{F}_2(\zeta)$ (as $4 \nmid 2 \cdot 3^k$); the translate lives in the degree-$4 \cdot 3^k$ compositum $\mathbb{F}_{2^{4h}}$, $h = 3^k$. (June 2026 pass; probe `experiments/exception_column_m4.py`.)

Proposition 12 (the corrected norm)**.** *[***proved***] Let* $\sigma = \mathrm{Frob}^{2h}$ *generate* $\mathrm{Gal}(\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}})$*. Then* $\sigma(4) = 5$*, and with* $\omega := \zeta^h = \kappa_2$*,*

$$N := \mathrm{N}_{\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}}}(\kappa_{3^k} + 4) = (\kappa + 4)(\kappa + 5) = \kappa^2 + \kappa + \omega.$$

Proof. $\mathbb{F}_{16} \cap \mathbb{F}_{2^{2h}} = \mathbb{F}_4$ because $\gcd(4, 2 \cdot 3^k) = 2$, so $\sigma|_{\mathbb{F}_{16}}$ is the nontrivial element of $\mathrm{Gal}(\mathbb{F}_{16}/\mathbb{F}_4)$, which swaps the two roots $4, 5$ of the Artin–Schreier minimal polynomial $y^2 + y + \omega$ of 4 over $\mathbb{F}_4$. (Equivalently: $2h \equiv 2 \pmod 4$ and the Frobenius orbit of 4 is $4 \to 6 \to 5 \to 7$, so $\sigma(4) = 4^{2^2} = 5$.) Then $(\kappa+4)(\kappa+5) = \kappa^2+\kappa+(4\otimes 5)$ and $4\otimes 5 = 4^2+4 = \omega$ by the same Artin–Schreier relation. □

Remark 3 (erratum)**.** The 2026-06-10 draft of this note stated the norm as $(\kappa+4)(\kappa+\mathrm{Frob}^2(4)) = (\kappa + 4)(\kappa + 6)$. The conjugate $6 = 4^2$ is $\mathrm{Frob}^1(4)$, one squaring short of $\mathrm{Frob}^2(4) = 5$. The slip is visible without computation: $4 \otimes 6 = 14 \notin \mathbb{F}_4$, so $(\kappa + 4)(\kappa + 6) \notin \mathbb{F}_{2^{2h}}$ and cannot be a norm.

The corrected norm collapses the $m = 4$ test into the *same* sparse trinomial model $\mathbb{F}_2[x]/(x^{2h} + x^h + 1)$ as the rest of the 3-power analysis: no compositum arithmetic is needed.

Corollary 2 (in-field criterion)**.** *[***proved***] For a prime* p *with* $f(p) = 2 \cdot 3^k$ *and* $v_p(2^{2h} - 1) = 1$*, write* $\overline{N} = N^{2^h}$ *and* $M_k := \overline{N}/N = N^{2^h-1} \in U$*. Then*

$$m_p = 4 \iff p \mid \mathrm{ord}(M_k) \iff M_k^{(2^h+1)/p} \neq 1.$$

Proof. $m_p \geq 4$ is Proposition 8, so $m_p = 4$ iff $\kappa + 4$ has no p-th root. By Propositions 1 and 2 with $E = 4h$, $f = 2h$ (and $v_p(2^{4h} - 1) = v_p(2^{2h} - 1) = 1$ by lifting the exponent, since $p \nmid 2$), that holds iff $N^{(2^{2h}-1)/p} \neq 1$. Since $\mathbb{F}_{2^{2h}}^\times = L^\times \times U$ with coprime orders, $p \mid 2^h + 1$, and $(2^{2h} - 1)/p = (2^h - 1) \cdot \big((2^h + 1)/p\big)$, the test equals $M_k^{(2^h+1)/p} \neq 1$. □

Conjecture 3 (D_k: the column analogue of C_k)**.** *The prime-to-*3 *part of* $\mathrm{ord}(M_k)$ *is full:* $\big((2^h + 1)/3^{k+1}\big) \mid \mathrm{ord}(M_k)$*. Equivalently (given the per-prime squarefreeness audits):* $m_p = 4$ *for every prime* p *with* $f(p) = 2 \cdot 3^k$*.*

[**certified** $k \leq 6$] D_k holds for $k = 2, \ldots, 6$: every prime factor of the fully factored $\Phi_{2\cdot 3^k}(2)$ divides $\mathrm{ord}(M_k)$. Hence $m_p = 4$ exactly — not merely ≥ 4 — for every prime p with $f(p) \in \{18, 54, 162, 486, 1458\}$:

f	primes p with $m_p = 4$
18	19
54	87211
162	163, 135433, 272010961
486	1459, 139483, 10429407431911334611, 918125051602568899753
1458	227862073, 3110690934667, 216892513252489863991753, 1102099161075964924744009, P78

(P78 is the certified 78-digit prime of $\Phi_{2\cdot 3^6}(2)$; PRP-local here, factordb marks it proven.) The anchor rows $m_{19} = m_{163} = m_{1459} = 4$ (DiMuro / calculator) are reproduced, never assumed; the remaining eleven rows are new. Independent cross-checks: the term algebra of `experiments/ordinal_excess_probe.py` re-derives the $p = 19$ root pattern ($m = 0, \ldots, 3$ root, $m = 4$ no root) and $m_{87211} = 4$; an explicit Artin–Schreier compositum model $\mathbb{F}[y]/(y^2 + y + \omega)$ verifies $\sigma(4) = 5$, the norm identity, and the full direct power test $(\kappa+4)^{(2^{4h}-1)/p} \neq 1$ on the smallest prime of each level $k \leq 6$.

[consistent] $k = 7, 8$: every *known* prime factor of $\Phi_{2\cdot 3^7}(2)$ and $\Phi_{2\cdot 3^8}(2)$ — seven and five primes respectively; factordb CF entries verified locally by divisibility, primality, and order tests, the small ones re-derived by sieve, and the list cross-audited against the raw factordb API after an adversarial review caught a dropped 43-digit $k = 7$ prime — also has $m_p = 4$. An $m_p \geq 5$ example, if one exists, now hides strictly inside the unfactored cofactors.

Proposition 13 (twisted norm tower)**.** *[**proved**] With $\eta = \zeta^3 = \kappa_{3^{k-1}}$,*

$$\mathrm{N}_{\mathbb{F}_{2^{2\cdot 3^k}}/\mathbb{F}_{2^{2\cdot 3^{k-1}}}}(N_k) = \eta^2 + \omega^2\eta + 1,$$

which is neither $N_{k-1} = \eta^2 + \eta + \omega$, nor one of its Frobenius conjugates (their exponent pattern is $(2^{j+1}, 2^j)$), nor a scaling $N_{k-1}(\lambda\eta)$ with $\lambda \in \mu_3$.

Proof. The generator τ of the degree-3 step sends $\zeta \mapsto \zeta^c$ with $c = 2^{2\cdot 3^{k-1}}$ a primitive cube root of unity modulo 3^{k+1}; the cube roots are $1 + a3^k$, so $\tau(\zeta) = \omega^a\zeta$ $(a \neq 0)$ while $\tau(\omega) = \omega$. Hence the norm is $\prod_{r^3=\eta}(r^2 + r + \omega)$ over the three cube roots r of η. Writing α, β for the two roots of $T^2 + T + \omega$ (so $\alpha + \beta = 1$, $\alpha\beta = \omega$), each factor is $(r+\alpha)(r+\beta)$, and the product collapses by the resolvent

$$\prod_{r^3=\eta}(r^2 + r + \omega) = (\alpha^3 + \eta)(\beta^3 + \eta) = \eta^2 + (\alpha^3 + \beta^3)\eta + (\alpha\beta)^3 = \eta^2 + \omega^2\eta + 1,$$

since $\alpha^3 + \beta^3 = (\alpha+\beta)^3 + \alpha\beta(\alpha+\beta) = 1 + \omega = \omega^2$ and $(\alpha\beta)^3 = \omega^3 = 1$. Machine-checked at $k = 2, 3, 4$. $\square$

So, unlike the γ tower of Proposition 10, old-prime full-order parts do *not* propagate up the column, and D_k is genuinely per-level: the certification at each k stands on its own.

Remark 4 (the 3-part)**.** **[consistent]** The 3-part of $\mathrm{ord}(M_k)$ carries no column information but is strikingly regular in range: $v_3(\mathrm{ord}\, M_k) = k+1$ (full) for odd k and $k-1$ (deficiency exactly 3^2) for even k, for $2 \leq k \leq 6$. No explanation is on record.

5 Boundedness

[conjectured] (Lenstra [5]) m_p is globally bounded. Lenstra gave unconditional lower-bound rules (singleton odd $\mathcal{Q}$ forces positive excess; $f(p) = 2\cdot 3^k$ forces excess at least 4) but left absolute boundedness open.

[proved] If the 0/1/4 candidate holds, then $m_p \leq 4$ for all p; proving Conjecture 2 would settle boundedness.

[proved] The 3-power column cannot produce $m \in \{2, 3\}$ (Proposition 9); any C_k failure jumps directly to $m_r \geq 4$.

[proved] (conditional on C_k) For every prime r with $f(r) = 3^j$, $j \leq k$: C_k implies $m_r = 1$. With C_k certified for $k \leq 6$, all primes in the 3-power column with $f(r) \leq 6561$ have $m_r = 1$ unconditionally.

[certified $k \leq 6$] (June 2026 pass) On the $2\cdot 3^k$ exception column the matching upper bound is now certified at every visible prime: $m_p = 4$ exactly for every prime with $f(p) = 2\cdot 3^k$, $k \leq 6$, and for every known prime at $k = 7, 8$ (Section 4.2). Boundedness on the exception column therefore holds unconditionally through $f = 1458$ and is equivalent to D_k (Conjecture 3) beyond.

[open] Boundedness outside the 3-power and $2 \cdot 3^k$ columns. The non-cyclotomic singleton chains (e.g. the 11-chain and the 23, 29, 47 components) have $m = 1$ rows in the calculator data and OEIS, but no order-criterion proof covers them the way C_k covers the 3-power column.

[open] Whether any prime p has $m_p \geq 5$. No counterexample is known; the question is not known to be decidable by any current structural argument outside the 3-adic columns.

6 Ranked next moves

1. *Dress rehearsal on m_{89}, m_{179}, m_{359} (highest priority).* Certify by direct order tests the calculator rows on which a m_{719} certificate depends. The component fields have degrees $E = 19{,}580$ ($p = 89$) and $E = 3{,}504{,}820$ ($p = 359$ – a stretch: hours to a day with `gf2x`-style arithmetic). This calibrates the cost model for the next move on the same code path and settles the provenance question ("what evidence is acceptable") before any new α ships. Confirm $m_{359} = 1$ first; the norm recurrence for $p = 719$ shifts if not. *Feasible now; blocks the next move.*

2. *Certify m_{719} via the Kummer-tower Frobenius norm.* Implement the binary-splitting norm $N_{2k} = N_k \cdot \mathrm{Frob}^{359k}(N_k)$ in the tower-sparse representation; measure the cost model on the calibrated path; if tower-aware Frobenius brings the $\sim$44 modular compositions to feasible size, run to certify or falsify $m_{719} = 1$. Factor out the shared `relative_norm_over_frobenius_orbit` abstraction (Remark 2). *Blocked on move 1 for the cost model.*

3. *[Done, 2026-06-12.] $m_p = 4$ exactly on the $2 \cdot 3^k$ column.* Settled at every prime current factor tables reach (Section 4.2, `experiments/exception_column_m4.py`): the corrected norm $(\kappa+4)(\kappa+5) = \kappa^2+\kappa+\omega$ collapses the test in-field; certified universally for $k \leq 6$, consistent at $k = 7, 8$. Successor moves: prove D_k (Conjecture 3) — the twisted norm tower (Proposition 13) shows why γ-style propagation is unavailable; factor the $\Phi_{2\cdot 3^7}(2)$ and $\Phi_{2\cdot 3^8}(2)$ cofactors (joins the ECM/GNFS batch of the next move); explain the 3-part parity (Remark 4).

4. *Convert C_7, C_8 from CONSISTENT to CERTIFIED.* ECM/GNFS jobs on the cofactors of $\Phi_{3^7}(2)$ and $\Phi_{3^8}(2)$; full factorization also emits a batch of new A380496 rows. *Background job; uncertain outcome; blocks nothing.*

5. *Exact-order census on the non-cyclotomic singleton chains.* Compute $\mathrm{ord}(\kappa_q + m)$ for small m in the explicit component fields for the 11-chain and the 23, 29, 47 components (degrees up to $\sim$5060; trivial locally); look for an analogue of the $L^\times \times U$ splitting and for any forced-translate mechanism. This is the main structural gap for boundedness outside the 3-power columns. *Feasible now; no structural theory yet.*

7 Provenance and validation

7.1 Independent oracle

`experiments/ordinal_excess_probe.py` is a small local term-algebra oracle – not a replacement for CGSuite [8] or the C++ calculator [9], but a way to verify the first subtle cases without using the Rust production tower as its own oracle. It implements the impartial term algebra used by the calculators, a multiplicative-order test for small component fields, and a fixed-base exponentiation path (ported from the calculator's strategy) for targeted root tests where full order factorization is unnecessary. Current probe output:

p	m	$\mathcal{Q}$	root?
7	0	$\{3\}$	yes
7	1	$\{3\}$	no
19	1	$\{9\}$	yes
19	4	$\{9\}$	no
73	1	$\{9\}$	no
47	1	$\{23\}$	no

The last line certifies $m_{47} = 1$ using only lower verified rows; since $f(47) = 23$ and $\mathcal{Q}(23) = \{23\}$, the carry is $\alpha_{47} = \kappa_{23} + 1 = \omega^{(\omega^7)} + 1$, the value now implemented in `tower.rs`.

7.2 The α_{47} promotion

The shipped state: `src/scalar/big/ordinal/tower.rs` carries α_{47} with the landmark test `locally_verified_alpha_47_landmark` and moves the refusal boundary to α_{53}; the documented boundaries in `nim.rs`, `mod.rs`, `src/games/nimber_game.rs`, `README.md`, and `OPEN.md` were updated in step. The full gate ran clean: `cargo fmt --check`, `cargo test`, `cargo check/clippy` with and without the `python` feature (warnings denied), and the probe under `py_compile`. Focused tests: `locally_verified_alpha_47_landmark` and `boundary_returns_none_past_prime_47`.

7.3 Supporting probes

The 3^k-family certification chain is `experiments/cyclotomic_3k_family.py` (committed, maintained), joined 2026-06-12 by `experiments/exception_column_m4.py` (the $2{\cdot}3^k$ exception column, Section 4.2; committed, maintained, stdlib-only, $\sim$2 minutes), with the research-run probes under `experiments/excess/` (machine-generated, rescued from ephemeral storage; triage before citing). The analysis in Sections 2–5 synthesizes the 2026-06-10 parallel research run plus the 2026-06-12 exception-column pass; each claim above carries its own tag, and no claim depends on an unverified solver.

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References

[1] J. H. Conway, *On Numbers and Games*, Academic Press, 1976.

[2] L. De Feo, H. Randriam, É. Rousseau, *Standard lattices of compatibly embedded finite fields*, Proc. ISSAC 2019; arXiv:1906.00870.

[3] J. DiMuro, *On* On$_p$, arXiv:1108.0962.

[4] E. Kaltofen, V. Shoup, *Subquadratic-time factoring of polynomials over finite fields*, Math. Comp. **67** (1998), 1179–1197.

[5] H. W. Lenstra, Jr., *On the algebraic closure of two*, Indag. Math. **39** (1977), 389–396; https://pub.math.leidenuniv.nl/~lenstrahw/PUBLICATIONS/1977e/art.pdf.

[6] H. W. Lenstra, Jr., *Nim multiplication*, Séminaire de Théorie des Nombres de Bordeaux, 1977–1978.

[7] OEIS Foundation Inc., entry A380496, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org/A380496.

[8] A. N. Siegel, *CGSuite* (software), NimFieldCalculator, https://github.com/aaron-siegel/cgsuite.

[9] D. Peeters, *transfinite-nim-calculator* (software), https://github.com/DjangoPeeters/transfinite-nim-calculator.

[10] J. von zur Gathen, V. Shoup, *Computing Frobenius maps and factoring polynomials*, Comput. Complexity **2** (1992), 187–224.